Exercise Set 14: Characteristic Classes and Chern-Weil Theory

Characteristic classes are cohomology classes associated to vector bundles they are the obstruction to trivializing a bundle. Chern-Weil theory gives a concrete way to compute them: take a connection on the bundle, compute its curvature F, and apply an invariant polynomial to get a closed differential form whose cohomology class is the characteristic class. This is the deepest bridge between local geometry (curvature) and global topology (cohomology). The Dirac monopole of Set 12 is the first Chern class. The Euler characteristic of Set 6B is the Euler class. The Witten index of Set 10 is the index of the Dirac operator. All of these are characteristic classes.

Recall of Definitions and Examples

R1. Write the definition of a **characteristic class** of a vector bundle $E \to M$. What two properties must it satisfy:

(a) **Naturality**: for any smooth map $f : N \to M$, $c(f^*E) = f^*c(E)$

(b) **Well-definedness**: $c(E)$ depends only on the isomorphism class of E, not on any choice of connection

R2. State the **Chern-Weil theorem**: let $E \to M$ be a complex vector bundle with connection ∇ and curvature F. For any Ad-invariant polynomial P on $\mathfrak{gl}(r, \mathbb{C})$, the form $P(F)$ is:

(a) Closed: $d(P(F)) = 0$

(b) Its cohomology class $[P(F)] \in H^*_{dR}(M)$ is independent of the choice of connection

This gives a map (the **Chern-Weil homomorphism**): $\mathrm{Inv}(\mathfrak{g}) \to H^*_{dR}(M)$.

R3. The **Chern classes** $c_k(E) \in H^{2k}(M; \mathbb{Z})$ of a complex vector bundle E of rank r are defined by:

$$\det\left(I + \frac{iF}{2\pi}\right) = \sum_{k=0}^{r} c_k(E) = 1 + c_1(E) + c_2(E) + \cdots + c_r(E).$$

Write $c_1(E)$, $c_2(E)$ explicitly in terms of F.

R4. The **Pontryagin classes** $p_k(E) \in H^{4k}(M; \mathbb{Z})$ of a real vector bundle E are defined by:

$$p_k(E) = (-1)^k c_{2k}(E_{\mathbb{C}})$$

where $E_{\mathbb{C}} = E \otimes \mathbb{C}$ is the complexification.

R5. The **Euler class** $e(E) \in H^n(M;\mathbb{Z})$ of an oriented real vector bundle of rank n satisfies: $e(E) \cup e(E) = p_{n/2}(E)$ (for even n). For the tangent bundle TM: $\int_M e(TM) = \chi(M)$.

R6. State the **Whitney sum formula**: $c(E \oplus F) = c(E) \cup c(F)$ and the **splitting principle** (every bundle formally splits into line bundles for the purpose of computing characteristic classes).

Theory

T1. (Chern-Weil: $d(P(F)) = 0$.)

Let ∇ be a connection on $E \to M$ with curvature $F = dA + A \wedge A$.

(a) Show that for the trace: $d(\mathrm{Tr}(F)) = 0$.
Hint: use the Bianchi identity $dF = -[A, F] = -(AF - FA)$ and the cyclic property of trace.

(b) Show that $d(\mathrm{Tr}(F \wedge F)) = 0$.

(c) For a general Ad-invariant polynomial P: Ad-invariance means $P(gFg^{-1}) = P(F)$. Differentiating: $P([X, F]) = 0$ for any matrix X. Use this to show $d(P(F)) = 0$.

T2. (Independence of connection.)

Let ∇_0 and ∇_1 be two connections on $E \to M$, with curvatures F_0 and F_1. Define the interpolating connection $\nabla_t = (1-t)\nabla_0 + t\nabla_1$.

(a) Show $P(F_1) - P(F_0) = d(\mathrm{CS}(A_0, A_1))$ where $\mathrm{CS}(A_0, A_1)$ is the **Chern-Simons form**:

$$\mathrm{CS}(A_0, A_1) = \int_0^1 P'\left(F_t, \frac{dA_t}{dt}\right) dt.$$

(b) Conclude: $[P(F_0)] = [P(F_1)]$ in $H^*_{dR}(M)$. The cohomology class is connection-independent.

(c) The Chern-Simons form is the transgression of $P(F)$ a form on M whose exterior derivative is the difference of two characteristic forms. This is exactly the Chern-Simons action from Nawata HW12.

T3. (Chern classes of $\mathbb{CP}^n$ Nawata HW12 P1.)

The tangent bundle $T\mathbb{CP}^n$ satisfies: $T\mathbb{CP}^n \oplus \epsilon \cong L^* \oplus \cdots \oplus L^*$ ($n+1$ copies) where L is the Hopf line bundle and ϵ is the trivial bundle.

(a) Since $c(\epsilon) = 1$ and $c(L^*) = 1 + x$ where $x \in H^2(\mathbb{CP}^n)$ is the generator, use the Whitney sum formula to show:

$$c(T\mathbb{CP}^n) = (1+x)^{n+1} \pmod{x^{n+1}}.$$

(b) Compute $c_k(T\mathbb{CP}^n) = \binom{n+1}{k} x^k$.

(c) Compute the total Chern class of $\mathbb{CP}^1$, $\mathbb{CP}^2$, $\mathbb{CP}^3$.

(d) For $\mathbb{CP}^1 \cong S^2$: show $c_1(T\mathbb{CP}^1) = 2x$ and $\int_{\mathbb{CP}^1} c_1 = 2 = \chi(S^2)$. ✓

Computation

C1. (First Chern class of the Hopf bundle Nawata HW12 P1.)

From Set 12 (C3), the Hopf bundle over S^2 has curvature: $F = -\frac{i}{2} \sin\theta \, d\theta \wedge d\varphi$.

(a) Compute $c_1 = \frac{i}{2\pi} F$.

(b) Compute $\int_{S^2} c_1$. Verify $c_1 = 1$ the first Chern number of the Hopf bundle.

(c) For the bundle with transition function $e^{in\varphi}$ (degree n): what is c_1?

(d) Show: all $U(1)$-bundles over S^2 are classified by $c_1 \in \mathbb{Z}$. This is the statement from Set 12.

C2. (Chern classes and the Euler characteristic.)

(a) For a complex manifold M of complex dimension n: show $\chi(M) = \int_M c_n(TM)$. *This is the Gauss-Bonnet-Chern theorem a generalisation of the 2D Gauss-Bonnet to all dimensions.*

(b) Verify for $\mathbb{CP}^1 = S^2$: $\int_{\mathbb{CP}^1} c_1(T\mathbb{CP}^1) = \int_{S^2} 2x = 2 = \chi(S^2)$. ✓

(c) Verify for $\mathbb{CP}^2$: $c_2(T\mathbb{CP}^2) = \binom{3}{2} x^2 = 3x^2$, $\int_{\mathbb{CP}^2} c_2 = 3 = \chi(\mathbb{CP}^2)$. ✓

C3. (The Pontryagin number and the signature.)

(a) For an oriented 4-manifold M: the first Pontryagin number $p_1(M) = \int_M p_1(TM)$ and the signature $\sigma(M)$ are related by the **Hirzebruch signature theorem**: $\sigma(M) = \frac{p_1(M)}{3}$.

(b) For $\mathbb{CP}^2$: $p_1(T\mathbb{CP}^2) = 3x^2$ (where $x \in H^2(\mathbb{CP}^2)$ with $\int x^2 = 1$), so $p_1 = 3$. Verify $\sigma(\mathbb{CP}^2) = 1 = p_1/3$. ✓

(c) For $\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$ (connected sum with reversed orientation): $\sigma = 1 + (-1) = 0$ and $\chi = 3 + 3 - 2 = 4$. Compute p_1 and verify the signature theorem.

C4. (The Chern-Simons form Nawata HW12 P3.)

For a $U(1)$-bundle on M^3 with connection A: the Chern-Simons form is $\mathrm{CS}(A) = A \wedge dA$ (abelian case).

(a) Show $d(\mathrm{CS}(A)) = dA \wedge dA = F \wedge F$.

(b) For the non-abelian case (Nawata HW12 P3): $\mathrm{CS}(A) = \mathrm{Tr}(A \wedge dA + \frac{2}{3} A \wedge A \wedge A)$. Show $d(\mathrm{CS}(A)) = \mathrm{Tr}(F \wedge F)$.

(c) The Chern-Simons action is $S_{CS} = \frac{k}{4\pi}\int_M \mathrm{CS}(A)$. If $M = \partial B$ for a 4-manifold B: show $S_{CS} = \frac{k}{4\pi}\int_B \mathrm{Tr}(F \wedge F) = k\pi p_1(B)$. Level quantization $k \in \mathbb{Z}$ follows from $p_1(B) \in \mathbb{Z}$.

Physics: Anomalies and Characteristic Classes

Characteristic classes appear in physics as ***anomalies****: classical symmetries that are broken by quantum effects. An anomaly is detected by a characteristic class of the gauge bundle. If the anomaly is non-zero, the theory is inconsistent. Cancelling anomalies imposes constraints on the gauge group and the matter content it selects which theories can exist. The cancellation of anomalies in the Standard Model is one of the most striking uses of characteristic classes in physics.*

P1. (The first Chern class as an anomaly.)

Consider a $U(1)$ gauge theory in 2D with action $S = \int_\Sigma A \wedge \star A$. Under a gauge transformation $A \to A + d\lambda$:

(a) Show S is gauge-invariant.

(b) The quantum partition function includes a path integral over fermion fields. For a chiral fermion (left-moving only), the fermion measure $\mathcal{D}\psi$ is NOT gauge-invariant. The anomaly is:

$$\delta_\lambda \log Z = \frac{i}{2\pi}\int_\Sigma \lambda\, dA = \frac{i}{2\pi}\int_\Sigma \lambda \cdot F.$$

This is proportional to c_1 of the gauge bundle.

(c) For $\Sigma = S^2$ and the Hopf bundle ($c_1 = 1$): the anomaly is nonzero the chiral fermion is inconsistent. What is the fix? (Add an opposite-chirality fermion to cancel the anomaly the theory must be non-chiral in 2D.)

P2. (Anomaly cancellation in 4D the Standard Model.)

In 4D, the gauge anomaly of a chiral fermion transforming under gauge group G is proportional to:

$$\mathcal{A} \propto \mathrm{Tr}[T_a\{T_b, T_c\}]$$

where T_a are the generators of G acting on the fermion.

(a) In the Standard Model with gauge group $U(1) \times SU(2) \times SU(3)$, there are 6 independent anomaly conditions. The quark charges (in units of $e/3$: up quark $+2$, down quark -1) and lepton charges (-3 for electron, 0 for neutrino) are fixed by anomaly cancellation. Write the $U(1)^3$ anomaly condition: $\sum_{\text{left}} q^3 - \sum_{\text{right}} q^3 = 0$ and verify it holds for one generation of quarks and leptons.

(b) The fact that quarks come in 3 colors (from $SU(3)$) and have charges $\pm 2/3, \mp 1/3$ while the electron has charge -1 is not a coincidence it is forced by anomaly cancellation. Explain: why does the existence of the top quark (discovered 1995) require the existence of the bottom quark?

(c) **The key point.** Characteristic classes (c_1, c_2, etc.) of the gauge bundle appear as anomaly coefficients. Anomaly cancellation = vanishing of certain characteristic classes. This is topology constraining the matter content of the universe.

P3. (The Dirac quantization revisited as a characteristic class.)

From Set 12: the magnetic monopole has first Chern number $c_1 = g/2\pi \in \mathbb{Z}$.

(a) The Chern-Weil formula: $c_1 = \frac{i}{2\pi} F$. Integrating: $\int_{S^2} c_1 = \frac{i}{2\pi} \int_{S^2} F \in \mathbb{Z}$. This is both the topological classification of bundles and the Dirac quantization condition.

(b) For the non-abelian monopole (Prasad-Sommerfield, Nawata HW11): the relevant class is p_1 of the $SU(2)$ bundle. The monopole charge is the Pontryagin number.

(c) The BPST instanton (Nawata HW11): the instanton number is $k = \frac{1}{8\pi^2} \int_{\mathbb{R}^4} \mathrm{Tr}(F \wedge F) \in \mathbb{Z}$ the second Chern number c_2 of the $SU(2)$ bundle. This is why instantons are topological they are classified by an integer.

P4. (The running theme: topology $\to$ physics.)

Characteristic class	Physical manifestation	Integer invariant
$c_1 \in H^2$	Dirac monopole charge	$\int_{S^2} c_1 \in \mathbb{Z}$
$c_1 \in H^2$	Quantum Hall conductance	$\sigma_{xy} = n\frac{e^2}{h}$
$c_2 \in H^4$	BPST instanton number	$k \in \mathbb{Z}$
$p_1 \in H^4$	Gravitational anomaly	$\sigma(M) \in \mathbb{Z}$
$e(TM)$	Euler characteristic	$\chi(M) \in \mathbb{Z}$

(a) The **quantum Hall effect**: the transverse conductance σ_{xy} of a 2D electron system is quantized: $\sigma_{xy} = n\frac{e^2}{h}$ where $n = \int_{T^2} c_1(L) \in \mathbb{Z}$ is the Chern number of the occupied band bundle over the Brillouin zone torus T^2. This is topology protecting a physical observable σ_{xy} cannot change unless a phase transition occurs.

(b) In every example in the table, an integer-valued characteristic class forces a physical quantity to be quantized. Write the general principle in one sentence.

(c) **Preview of Set 17 (Index theory.)** The Atiyah-Singer index theorem expresses the index of a differential operator (an analytic quantity) as an integral of characteristic classes (a topological quantity):

$$\mathrm{index}(D) = \int_M \hat{A}(TM) \wedge \mathrm{ch}(E).$$

This is the grand unification of Gauss-Bonnet, Hirzebruch, Riemann-Roch, and the Witten index all as special cases of one formula.

In physics, characteristic classes appear as anomaly coefficients, instanton numbers, and topological charges. The quantization of these physical quantities is a direct consequence of the integrality of characteristic classes. Anomaly cancellation in the Standard Model which determines which particles can exist is the condition that certain characteristic classes vanish. Topology, literally, selects the laws of physics.***Reflection.*** *Characteristic classes are the cohomological shadows of bundles: c_1 measures the twisting of a line bundle, c_2 measures the topology of an $SU(2)$ bundle, p_1 measures the Pontryagin number, $e(TM)$ gives the Euler characteristic. Chern-Weil theory computes them from curvature local data gives global information.*

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